Homework 2

Independence, random variables, and discrete laws

60 points; 10 questions. Questions 6 and 8 are worth 10 points each; the others are worth 5 points each.

Due: October 12, 2026, at 10 p.m. Pacific. Final no-penalty grace cutoff: October 13, 2026, at 10 p.m. Pacific.

Handwrite your work on paper or a tablet, subject to approved accommodations. Submit one legible PDF in Gradescope and map its pages to all ten questions. Show your reasoning, support, and assumptions. Exact expressions are accepted unless stated otherwise. Write independently and disclose collaborators. Do not use AI tools to generate, solve, check, or rewrite assessed work.

Notation. $I_{\{C\}}$ is one when condition C holds and zero otherwise. Indicator, complete piecewise, and explicit-support-with-zero-elsewhere PMFs are equally valid. Preserve integer restrictions. Use a piecewise form for a CDF when it is clearer.

Study sequence. Start Q1–3 after the October 5 lesson, Q4–7 after October 7, and Q8–10 after October 9. These are suggested stages, not additional deadlines.

1. Independence and disjointness

5 points

Let $P(A) = 2/5$ and $P(B) = 1/2$. Treat the two cases below separately.

(a) [2] Assume A and B are independent. Find $P(A \cap B)$, $P(A \cup B)$, and $P(A \mid B)$.

(b) [3] Instead assume A and B are disjoint. Find the same three probabilities and decide whether they are independent. Finally, is the claim "disjoint events can never be independent" correct? Explain what happens if one event has probability zero.

2. Testing independence in one experiment

5 points

Choose one of the twelve tickets numbered $1, 2, \ldots, 12$, with every ticket equally likely. Let A mean that its number is divisible by 3, let B mean that it is at most 6, and let C mean that it is at most 5.

(a) [2] List A, B, and $A \cap B$. Use probabilities to determine whether A and B are independent.

(b) [3] Find $P(A \mid C)$ and determine whether A and C are independent. Explain why events based on the same random ticket need not be dependent, referring to both comparisons.

3. Outcomes, indicators, and unequal weights

5 points

Toss a coin twice independently. Each toss has head probability $1/3$. Let X count the heads and let $I = I_{\{\text{the first toss is H}\}}$.

(a) [3] Make a table with one row for each ordered outcome. Include its probability, X, and I. Write the events $\{X = 1\}$ and $\{I = 1\}$ as sets of outcomes.

(b) [2] Find $P(X = 1 \mid I = 1)$ and $P(I = 1 \mid X = 1)$. Explain which outcomes remain under each condition and why counting those outcomes equally is valid in only one of these two calculations.

4. Normalizing and reading a compact PMF

5 points

A discrete random variable has PMF

$$p_X(x) = c(2x+1)I_{\{x\in\{0,1,2,3\}\}}.$$

The indicator is one on the four listed integers and zero elsewhere.

(a) [2] Find c and give the four probabilities. Check nonnegativity and normalization.

(b) [3] Find $p_X(1.5)$, $P(1 \le X < 3)$, and $F_X(1.5)$. Explain why the PMF value and CDF value at 1.5 differ. Would replacing the indicator by $I_{\{0\le x\le 3\}}$ still specify the same PMF? Explain.

5. A distribution built by sampling without replacement

5 points

A box contains five distinct red tokens and three distinct blue tokens. Choose a three-token subset uniformly from all such subsets, without replacement. Let X count the blue tokens selected.

(a) [3] Give the support and derive the complete PMF by counting subsets. Report the probabilities in a table and verify that they sum to one.

(b) [2] Find $P(X \ge 1)$ by summing appropriate masses and again by using a complement. Explain why the four possible values of X are not equally likely.

6. From a PMF to a CDF

10 points

The distribution of X is

x	-2	0	3	5
$p_X(x)$	$1/8$	$3/8$	$1/4$	$1/4$

and $p_X(x) = 0$ elsewhere.

(a) [6] Write $F_X(t)$ for every real t. Sketch it, showing open and filled endpoints at each jump.

(b) [4] Find $P(-2 < X \le 3)$, $P(-2 \le X < 3)$, and $P(X = 0)$. For each, identify the included support values or the relevant jump. State exactly which event has probability $F_X(3) - F_X(-2)$.

7. Recovering masses from a CDF

5 points

A random variable Y has CDF

$$G(t) = \begin{cases} 0, & t < -1, \\ 1/5, & -1 \le t < 2, \\ 3/5, & 2 \le t < 4, \\ 1, & t \ge 4. \end{cases}$$

(a) [2] Recover the support and complete PMF of Y. A student writes $P(Y = 2) = G(2) = 3/5$. Correct the statement and explain the error.

(b) [3] Find $P(-1 \le Y < 4)$ and $P(Y = 2 \mid Y > 0)$. Show the probabilities used in the conditional ratio. Explain why $G(4) - G(-1)$ does not equal the first requested probability.

8. From a pattern to a binomial count

10 points

A randomized test succeeds on each of five mutually independent runs with probability 2/5. Let N be the number of successes; write S for success and F for failure.

(a) [6] Find the probability of the specified pattern $SFSFF$. Derive the complete PMF of N by combining the probability of one pattern with the number of patterns. State the binomial assumptions.

(b) [4] Find $P(N \geq 1)$ and $P(N \geq 2)$ using complements. Identify the excluded outcomes in each calculation. Exact powers or fractions are sufficient.

9. When the binomial assumptions fail

5 points

For part (a), assess the stated trial mechanism; no replacement distribution needs to be named.

(a) [2] For each case, decide whether the binomial trial assumptions apply. If yes, give n, p; if not, identify a failed assumption.

(i) Count successes in eight independent trials, each with success probability 0.2.

(ii) Count successes in eight independent trials: the first four have probability 0.1 and the last four have probability 0.3.

(iii) Count red tokens in eight uniform draws without replacement from five red and fifteen blue tokens.

(iv) Count how many independent attempts, each with success probability 0.2, are needed to obtain the first success.

(b) [3] Two independent attempts have success probabilities 1/4 and 3/4, respectively. Let Z count their successes. Derive its complete PMF by considering the four patterns. A student proposes $Z \sim \text{Bin}(2, 1/2)$ because the average success probability is 1/2. Use your PMF to evaluate this claim.

10. Two different observations about the same system

5 points

Four mutually independent components each function with probability 2/3. Let N count the functioning components.

(a) [2] Find $P(N = 3 \mid N \geq 1)$. Show the numerator and denominator of the conditional probability.

(b) [3] Instead, suppose component 1 is known to function. Find the probability that exactly three components function. Explain why this condition differs from $N \geq 1$, identifying what is fixed and what remains random.